Technically savvy scientists and engineers have put much effort into conceiving far-future technologies that might make possible near-light-speed travel. You can learn a lot about their ideas by browsing the web. It will take many centuries for humans to make any of those ideas real, I think. But they do convince me that ultra-advanced civilizations are likely to travel between the stars at a tenth the speed of light or faster.

Here are three far-out examples of near-light-speed propulsion that intrigue me.

## Thermonuclear Fusion

Thermonuclear fusion is the most conventional of the three ideas. R&D to develop controlled-fusion power plants on Earth was initiated in the 1950s, and full success will not come until the 2050s. A full century of R&D! That's a realistic measure of the difficulties.

And what will fusion power plants in 2050 mean for spacecraft propulsion by fusion? The most practical designs may achieve 100 kilometers per second, and conceivably 300 kilometers per second by the end of this century. A whole new approach to harnessing fusion will be required for reaching near light speed.

A simple calculation shows fusion's possibility: When two deuterium (heavy hydrogen) atoms are fused to form a helium atom, 0.0064 (nearly 1 percent) of their rest mass gets converted into energy. If this were all transformed to kinetic energy (energy of motion) of the helium atom, the atom would move at about one-tenth the speed of light.[26] This suggests that, if we could convert all the fusion energy of deuterium fuel into ordered motion of a spacecraft, we could achieve a